DIAMETER: Protocol for AAA

Course objectives:

Since the era of modem pools and dial-up connections to the internet, backend authentication servers and tool had provided facilities to control secured access by supporting authentication and authorization.
Today's networks, specifically mobile networks and broadband access, had soon outgrown the rather simple capabilities provided by DIAMETER's predecessor, RADIUS.
Born out of IETF, DIAMETER was designed to be a multi-purpose Authentication, Authorization and Accounting protocol for many use cases and scenarios.

This training course aims at providing a solid foundation to understand the DIAMETER, its applications and the interaction with the backend infrastructure.
We will make us of the open-source DIAMETER implementation, named "freeDiameter", to provide examples and hands-on experience, supporting the course material.

Duration Instructional methods:

Option 1: On-Site Training: 3 days,
Classroom-style lecture using Power-point slides with interactive discussion,
Questions & Answers after each module.

Option 2: Virtual Classroom: 3 days total with 4 x 1.5 hours per day

Option 3: Virtual Classroom: 6 days total with 2 x 1.5 hours per day

Upon successful registration participants will receive dedicated LMS account credentials, allowing access to pre-training documents and eLearning material.
All participants will also receive comprehensive course materials.

Target Audience:

The course is addressed to technical and staff from the technology, marketing and management field, system architects and system designers, programmers, students and researchers, requiring a bigger picture of the DIAMETER protocol, it's concepts, architectures, standards, implementations, applications and operations

Course Contents:

- Introduction to DIAMETER
  - Purpose of AAA
  - Motivation for DIAMETER
  - Differences between RADIUS and DIAMETER
  - Introduction to "freeDiameter" open-source implementation

- DIAMETER architecture
  - DIAMETER nodes
  - DIAMETER protocol structure
  - DIAMETER applications
  - DIAMETER peers
  - DIAMETER Message format and message structure
  - Attribute-Value Pairs (AVP's)
  - DIAMETER sessions and transactions
  - DIAMETER agents (redirect, relay, proxy and translation agents)

- DIAMETER node-to-node communication
  - Peer connections and DIAMETER sessions
  - Peer Discovery: static vs. dynamic
  - Connection establishment
  - Capabilities exchange
  - Peer connection maintenance and recovery
  - TCP and SCTP Multi-Homing in DIAMETER
  - DIAMETER end-to-end communication
  - DIAMETER routing
  - Intra-Realm versus Inter-Realm Communication
  - Diameter Routing and Inter-Connection Networks
  - Overload handling

- Security functionality in DIAMETER
  - Overview: Unkeyed Primitives, Symmetric Key Primitives, Asymmetric Key Primitives
  - Security Threats
  - Security Services: Secure Transport and Authorization
  - PKI example in "freeDiameter" – protecting exchange via TLS

- DIAMETER applications
  - Accounting
    - Services, Records and Setup
    - Accounting AVP's
    - Accounting example with "freeDiameter"
    - Accounting example: 3GPP Rf Interface for Offline Charging
  - Credit Control
    - Credit Control Request-Answer model
    - Credit Control example: 3GPP Ro Interface for Online Charging
  - Quality of Service
    - Push Mode vs Pull Mode
    - Authorization methods
    - QoS Application Session Establishment and Management
    - Session Termination
  - Use Case: the 3GPP EPC S6a Interface
    - S6a Overview
    - Common commands and AVP's for S6a
    - Authentication on S6a
    - Location Management examples: Update Location, Cancel Location
    - Subscriber Data Handling examples: Insert Subscriber Data, Delete Subscriber Data
    - Notifications
    - Ending Subscriber Sessions – example: Purge-UE

- DIAMETER extensions
  - Overview of extension strategies
  - Extending existing AVP's
  - Creating new AVP's
  - Extending commands
  - Vendor-Specific AVP extensions

Pre-requisites:

A solid foundation about Internet protocols, mobile networks architecture, IP networks and experience with IT infrastructure will support the learning experience.

Course Code: DIAMETER01